

2009 年

$$(\beta_1 \beta_2 \beta_3) = (\alpha_1 \alpha_2 \alpha_3) P$$

$$\text{得 } P = (\alpha_1 \alpha_2 \alpha_3)^{-1} (\beta_1 \beta_2 \beta_3)$$

$$= \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & -1 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 2 & -1 \\ 3 & -2 & 0 \end{bmatrix}$$

$$(1) B = P^{-1}AP \quad \text{简易算法: } (\alpha_1, \alpha_2, \alpha_3 | \beta_1, \beta_2, \beta_3)$$

$$(2) \text{ 求 } Ax=0$$

$$A \sim \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \quad \alpha = (1, -1, 1)^T$$

$$\text{核 } \alpha = \alpha_1 - \alpha_2 + \alpha_3 = (1, 0, 1)^T$$

值域

$$\begin{aligned} R(f) &= \text{span} \{ f(\alpha_1), f(\alpha_2), f(\alpha_3) \} \\ &= \text{span} \{ \alpha_1 + \alpha_2 - \alpha_3, \alpha_2 + 2\alpha_3, -\alpha_1 + 3\alpha_3 \} \\ &= \text{span} \{ \alpha_1 + \alpha_2 - \alpha_3, \alpha_2 + 2\alpha_3 \} \end{aligned}$$

二. 解:

$$A = \begin{bmatrix} \frac{1}{2} & 0 & \frac{3}{2}i \\ 0 & 2 & 0 \\ -\frac{3}{2}i & 0 & \frac{1}{2} \end{bmatrix}$$

$$|A - \lambda E| = (2 - \lambda)(\lambda - 2)(\lambda + 1)$$

$$\lambda = 2 \text{ 时, } A - 2E \sim \begin{bmatrix} 1 & 0 & -i \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \alpha_1 = (0, 1, 0)^T$$

$$\alpha_2 = (i, 0, 1)^T$$

$$\downarrow$$

$$\alpha_3 = (-i, 0, 1)^T$$

$$\downarrow$$

$$\alpha_4 = (-\frac{3}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}})^T$$

$$\lambda = -1 \text{ 时, } A + E \sim \begin{bmatrix} 1 & 0 & i \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$U = (u_1, u_2, u_3) = \begin{bmatrix} 0 & \frac{i}{\sqrt{2}} & -\frac{i}{\sqrt{2}} \\ 1 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$\text{令 } X = UY, Y = (y_1, y_2, y_3) \text{ 则}$$

$$f(X) = X^H A X = Y^H U^H A U Y = 2\bar{y}_1 y_1 + 2\bar{y}_2 y_2 - \bar{y}_3 y_3$$

三. 解:

$$AA^H = \begin{bmatrix} 2 & 0 \\ 0 & -i \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & i & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

特征值为 $\lambda = 2, 1$

$$\text{当 } \lambda = 4 \text{ 时, } AA^H - 4E \sim \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \text{ 故 } u_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{当 } \lambda = 1 \text{ 时, } AA^H - E \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \text{ 故 } u_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\text{当 } \lambda = 0 \text{ 时, } AA^H \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \text{ 故 } u_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\text{则 } U = (u_1, u_2, u_3) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, U_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$\text{令 } V_1 = A^H U_1 \Delta^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & -i \end{bmatrix}$$

$$\text{则 } V = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$$

$$\text{此时, } A = U \begin{bmatrix} 2 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} V^H$$

$$\begin{aligned} \text{IV. } \textcircled{1} \|UA\|_2 &= \max_j (\lambda_i [(UA)^H UA])^{\frac{1}{2}} \\ &= \max_j (\lambda_i [A^H U^H UA])^{\frac{1}{2}} \\ &= \max_j (\lambda_i [A^H A])^{\frac{1}{2}} = \|A\|_2 \end{aligned}$$

$$\begin{aligned} \textcircled{2} \|AV\|_2 &= \max_j (\lambda_i [(AV)^H AV])^{\frac{1}{2}} \\ &= \max_j (\lambda_i [V^H A^H AV])^{\frac{1}{2}} \\ &= \max_j (\lambda_i [A^H A])^{\frac{1}{2}} = \|A\|_2 \end{aligned}$$

$$\textcircled{3} (UAV)^H (UAV) = V^H A^H U^H U A V = V^H A^H A V$$

$$\therefore \|UAV\|_2 = \max_j (\lambda_i [V^H A^H A V])^{\frac{1}{2}} = \max_j (\lambda_i [A^H A])^{\frac{1}{2}} = \|A\|_2$$

$$\text{从而 } \|UA\|_2 = \|AV\|_2 = \|UAV\|_2 = \|A\|_2$$

五.

$$P(A) = 10 - \sqrt{10}$$

$$\begin{aligned} \sum_{k=0}^{\infty} \frac{k+1}{10^{kH}} x^k &= \left[\sum_{k=0}^{\infty} \left(\frac{x}{10} \right)^{kH} \right]' \\ &= \left(\frac{\frac{x}{10}}{1 - \frac{x}{10}} \right)', \quad \left| \frac{x}{10} \right| < 1 \\ &= \frac{10}{(10-x)^2}, \quad |x| < 10 \end{aligned}$$

$$(1) P(A) < R \text{ 故 } \sum_{k=0}^{\infty} \frac{kH}{10^{kH}} A^k \text{ 绝对收敛.}$$

$$\begin{aligned} (2) \sum_{k=0}^{\infty} \frac{kH}{10^{kH}} A^k &= 10 \cdot (10E - A)^{-2} \\ &= 10 \cdot (\sqrt{10}E)^{-2} = E \end{aligned}$$

$$\text{六. (1) } A^H = \begin{bmatrix} 3 & 1 & 0 & 0 \\ 0 & 3 & 1 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix} \quad A \sim A^H$$

故有相同 Jordan 形.

$$\phi_A(\lambda) = (\lambda - 3)^3$$

$$p(\lambda) = a_0 + a_1 \lambda + a_2 \lambda^2$$

$$p(3) = a_0 + 3a_1 + 9a_2 = f(3)$$

$$p'(3) = a_1 + 6a_2 = f'(3)$$

$$p''(3) = 2a_2 = f''(3)$$

$$\sin \pi A = -6\pi E + 2\pi A$$

$$\omega_{\frac{\pi}{2}} A =$$

$$\therefore \begin{cases} a_0 = f(3) - 3f'(3) + \frac{9}{2}f''(3) \\ a_1 = f'(3) - 3f''(3) \\ a_2 = \frac{1}{2}f''(3) \end{cases}$$

$$(2) |A - \lambda E| = (2 - \lambda)(\lambda^2 - 1)$$

$$\lambda_1 = 1, \lambda_2 = -1, \lambda_3 = 2$$

$$p(\lambda) = a_0 + a_1\lambda + a_2\lambda^2$$

$$p(1) = f(1) = a_0 + a_1 + a_2$$

$$p(-1) = f(-1) = a_0 - a_1 + a_2$$

$$p(2) = f(2) = a_0 + 2a_1 + 4a_2$$

$$\therefore \begin{cases} a_0 = f(1) + \frac{5}{6}f(-1) - \frac{1}{3}f(2) \\ a_1 = \frac{1}{2}[f(1) - f(-1)] \\ a_2 = \frac{1}{6}f(1) - \frac{1}{3}f(-1) + \frac{1}{3}f(2) \end{cases}$$

代入

七. 证明:

$$\text{设 } A = PJP^{-1} = P \text{diag}(J_1, J_2, \dots, J_r)P^{-1}$$

$$\text{则 } e^{kA} = P \text{diag}(e^{kJ_1}, e^{kJ_2}, \dots, e^{kJ_r})P^{-1}$$

$$\text{则 } e^{kJ_i} = \begin{bmatrix} e^{k\lambda_i} & ke^{k\lambda_i} & \dots & \frac{k^{d_i-1}}{(d_i-1)!}e^{k\lambda_i} \\ & e^{k\lambda_i} & \dots & \frac{k^{d_i-2}}{(d_i-2)!}e^{k\lambda_i} \\ & & \ddots & \vdots \\ & & & e^{k\lambda_i} \end{bmatrix}_{d_i \times d_i}$$

$$\begin{aligned} |e^{kA}| &= |P| |\text{diag}(e^{kJ_1}, e^{kJ_2}, \dots, e^{kJ_r})| |P^{-1}| \\ &= |\text{diag}(e^{kJ_1}, e^{kJ_2}, \dots, e^{kJ_r})| \\ &= e^{kd_1\lambda_1} e^{kd_2\lambda_2} \dots e^{kd_r\lambda_r} \\ &= e^{k(d_1\lambda_1 + d_2\lambda_2 + \dots + d_r\lambda_r)} \\ &= e^{k \text{tr}(A)} \quad \text{得证} \end{aligned}$$

$$18. \frac{d}{dx} \left(\int_0^{x^2} A(t) dt \right) = 2x \cdot A(x^2)$$

$$= 2x \cdot \begin{bmatrix} e^{2x^2} & x^2 e^{x^2} & x^4 \\ e^{-x^2} & 2e^{2x^2} & 0 \\ 3x^2 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 2xe^{2x^2} & 2x^3 e^{x^2} & 2x^5 \\ 2xe^{-x^2} & 4xe^{2x^2} & 0 \\ 6x^3 & 0 & 0 \end{bmatrix}$$

$$19. Ax = b$$

$$A = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} \quad b = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

B C

$$\therefore A^+ = C^H (CC^H)^{-1} (B^H B)^+ B^H$$

$$x = A^+ b$$